

Load Interconnection Request System Impact Study Report

For Queue Project Q11-0072

Blythewood 100 kV
July 04, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0072. The Interconnection Customer proposes a 183.2 MW large load (data center) facility interconnecting at the Blythewood 100 kV substation in the Illinois area. The request was received on September 19, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 57 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,372,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: BLYTHEWOOD SUBSTATION / RAPIDS 100 KV Line 2
Contingency: BLYTHEWOOD / RAPIS 100 KV Circuit 1
Loading: 174.3% (Rating: 176 MVA)
DFAX: 68.5%
Pre-project loading: 103.1%
2. Facility: RAPIDS / BLYTHEWOOD 100 KV Circuit 1
Contingency: RAPIDS / BLYTHEWOOD 100 KV Line 2
Loading: 167.4% (Rating: 184 MVA)
DFAX: 68.5%
Pre-project loading: 99.1%
3. Facility: LAS GALLINAS 230/100kV Transformer Bank
Contingency: MAZOMANIE / NRG ROCKFORD I 230 KV
Loading: 107.7% (Rating: 221 MVA)
DFAX: 20.5%
Pre-project loading: 90.7%
4. Facility: MAZOMANIE / NRG ROCKFORD I 230 KV
Contingency: MAZOMANIE SUBSTATION / BYRON 230 KV
Loading: 103.8% (Rating: 641 MVA)
DFAX: 46.9%
Pre-project loading: 90.4%
5. Facility: RAPIDS / JUDICE 100 KV Line 1
Contingency: JUDICE / RAPIDS 100 KV Circuit 2
Loading: 102.0% (Rating: 151 MVA)
DFAX: 30.5%
Pre-project loading: 65.1%

6. Facility: JUDICE / RAPIDS 100 KV Ckt 2
 Contingency: JUDICE / RAPIDS 100 KV Cir. 1
 Loading: 102.0% (Rating: 151 MVA)
 DFAX: 30.5%
 Pre-project loading: 65.1%
7. Facility: BOCA RATON / MORRIS COGENERATION LLC SUBSTATION 230 KV
 Contingency: DRESDEN GENERATING STA / MORRIS COGENERATION LLC 230 KV line
 Loading: 101.5% (Rating: 579 MVA)
 DFAX: 5.3%
 Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLYTHEWOOD SUBSTATION / RAPIDS 100 KV Line 2	Rebuild line with higher-capacity conductor	\$2,186,000	100% (\$2,186,000)
RAPIDS / BLYTHEWOOD 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$2,271,000	100% (\$2,271,000)
LAS GALLINAS 230/100 kV Autotransformer	Replace transformer with higher-rated unit	\$5,676,000	82% (\$4,649,000)
MAZOMANIE / NRG ROCKFORD I 230 KV	Reconductor line	\$2,970,000	82% (\$2,451,000)
RAPIDS / JUDICE 100 KV Line 1	Reconductor line	\$950,000	100% (\$950,000)
JUDICE / RAPIDS 100 KV Circuit 2	Reconductor line	\$944,000	100% (\$944,000)
BOCA RATON / MORRIS COGENERATION LLC SUBSTATION 230 KV	Reconductor line	\$13,931,000	7% (\$921,000)
	Total allocated to project		\$14,372,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.